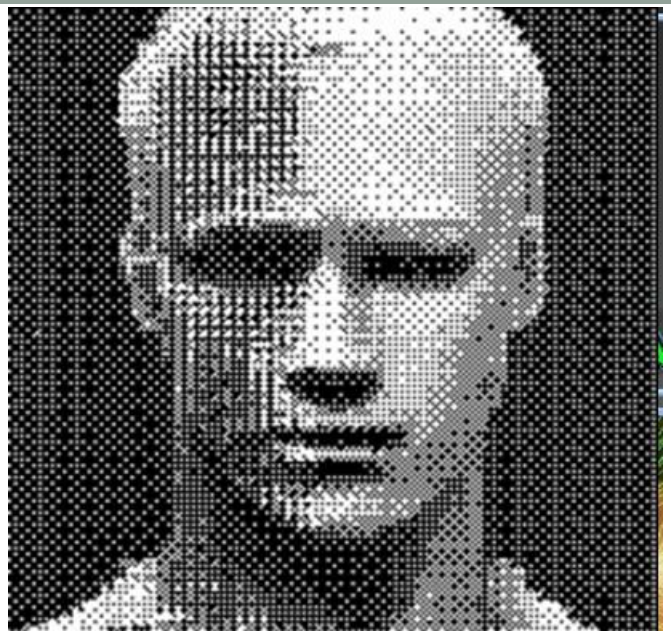


DIGITAL IMAGE PROCESSING

Dr. Manar Donia

CMPn332 - CMPN332 & CMP 432

Course objectives



The course discusses the fundamental concepts, aspects, and terminologies of digital image processing.



Exploring image fundamentals and mathematical transforms necessary for image processing and enhancement techniques in spatial and frequency domains. also, study the image compression and segmentation procedures.



It provides the basics of the OpenCV application programming interface that allows for developing programming skills.

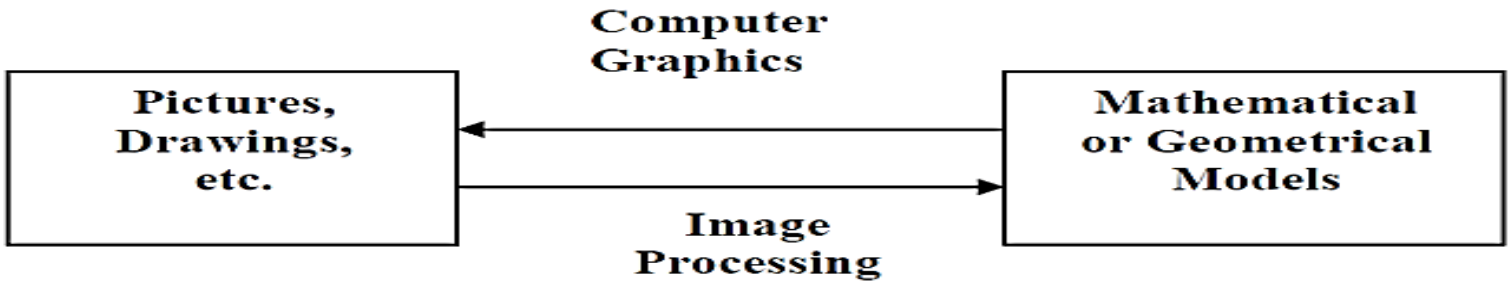
Course Assessment_2018 & 2020

Assessment Method	Timing	Grade
Semester Work: seminars, quizzes assignments and reports	**Bi-Weekly**	**20**
Mid-Term Exam	**8th Week**	**20**
Practical Exam	**13th Week**	**20**
Written Exam	**15th Week**	**40**
Total		**100**

Course Assessment_2012

Assessment Method	Timing	Grade
Semester Work: seminars, quizzes assignments and reports	**Bi-Weekly**	**10**
Mid-Term Exam	**7th Week**	**10**
Practical Exam	**12th Week**	**20**
Written Exam	**15th Week**	**60**
Total		**100**

Computer graphics vs Image processing





computer graphics are concerned with the creation of images



Image Processing: analyze pictures to derive descriptions (often in mathematical or geometrical forms) of objects appeared in the pictures.

What is the objective of image processing?

- **Image Analysis and Interpretation→** Extract meaningful information from an image for further analysis or decision-making by machines or humans.
- **Image Enhancement→** Increasing contrast, reducing noise, sharpening details, or adjusting brightness.
- **Image Restoration→** Restore image to its original or an improved state.
- **Image Compression→** Reduce the size of an image file for efficient storage or transmission without losing significant quality
- **Image Segmentation→** Divide an image into meaningful regions or segments to isolate areas of interest for further analysis.

What is the objective of image processing?

The ultimate goal of digital image processing is to build systems that can automatically analyze, interpret, enhance, and manipulate digital images such systems are:

- **Medical Imaging System**→ build a system that can detect tumors in medical scans.
- **Security and Surveillance System**→ Image processing is crucial for real-time monitoring and detection of suspicious activities in security systems.
- **Biometric Security System**→ Image processing algorithms are used to capture, enhance, and analyze fingerprints for identification purposes.
- **License Plate Recognition System**→ Recognizing vehicle license plates in traffic enforcement.

What's an Image?



- Image is a 2D projection of 3D object.
- So, image can be defined as 2D function F(x, y)
  - where x, y represent spatial coordinates
  - and the amplitude of F at any point is called the intensity at that point.

Image Types

- Analog Image and Digital Image



Analog Image→ continuous variation in tones



Digital Image→ finite set of digital values called pixels

What is digital image?

- A **digital image** is a representation of a scene, in a format that a computer can process.
- It consists of a grid of tiny units called **pixels** (short for "picture elements")
- Each pixel represents a specific color or intensity.
- The image is stored digitally, meaning it is made up of digital data (1s and 0s) that a computer can manipulate, display, or transmit.

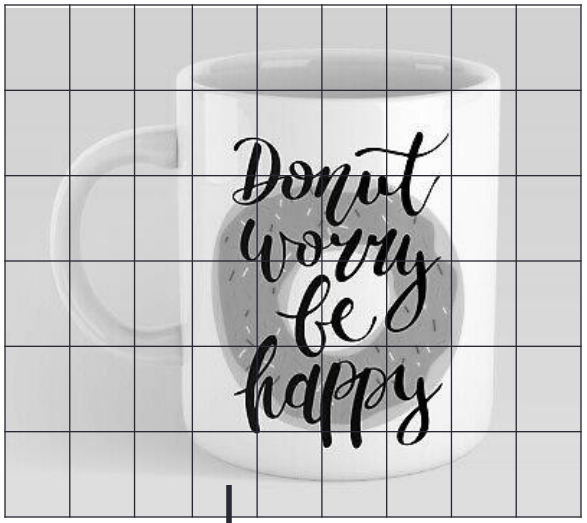
Digital Image color models

RGB



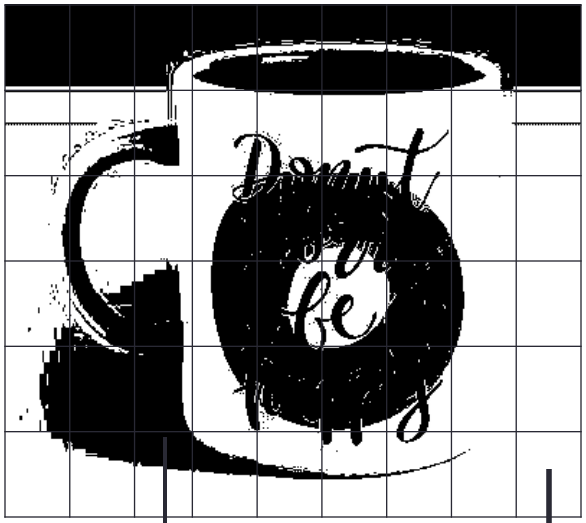
131,173,39

Grayscale



127

B&W



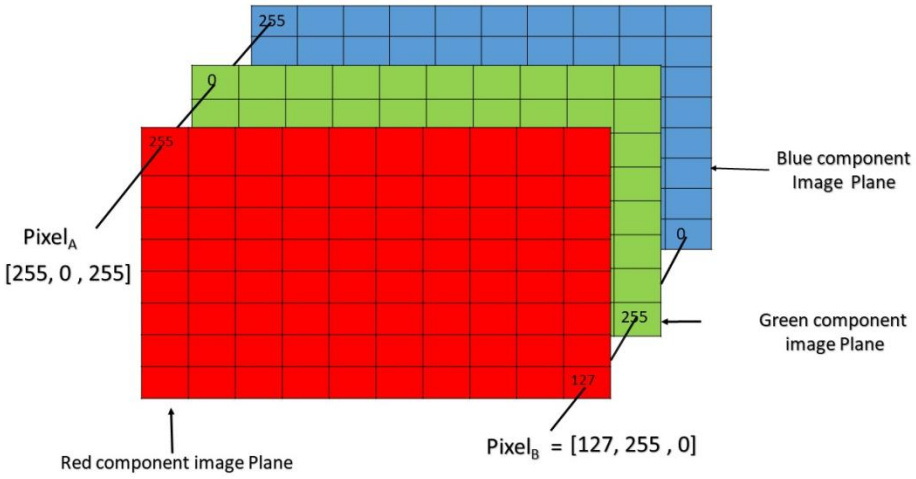
0

255

Digital image is composed of a finite number of pixels **"picture elements"**, each of which has a particular location and value

Digital Image color models

RGB



Grayscale

$$\begin{bmatrix} 52 & 55 & 61 & 66 & 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 & 109 & 85 & 69 & 72 \\ 62 & 59 & 68 & 113 & 144 & 104 & 66 & 73 \\ 63 & 58 & 71 & 122 & 154 & 106 & 70 & 69 \\ 67 & 61 & 68 & 104 & 126 & 88 & 68 & 70 \\ 79 & 65 & 60 & 70 & 77 & 68 & 58 & 75 \\ 85 & 71 & 64 & 59 & 55 & 61 & 65 & 83 \\ 87 & 79 & 69 & 68 & 65 & 76 & 78 & 94 \end{bmatrix}$$

In our work we mostly work on grayscale system as RGB image has 3D while grayscale has just one, so, everything tend to be more costly in RGB system

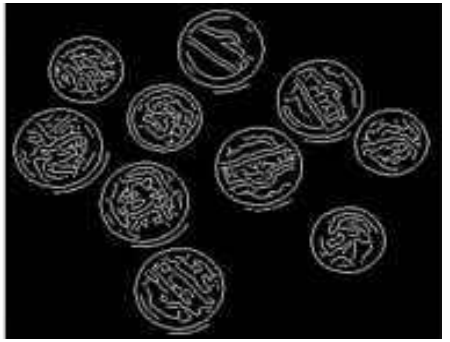
What type of processing can be done on digital image?





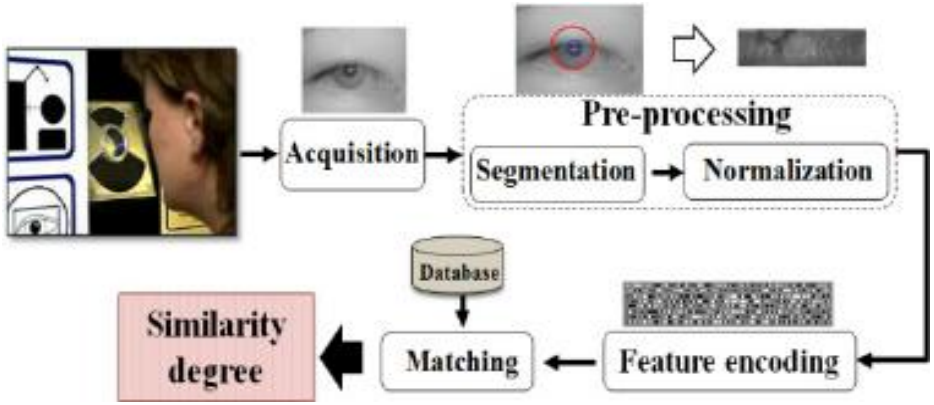
- Low level processing
  - Input image
  - Output image
    - Example: noise removal



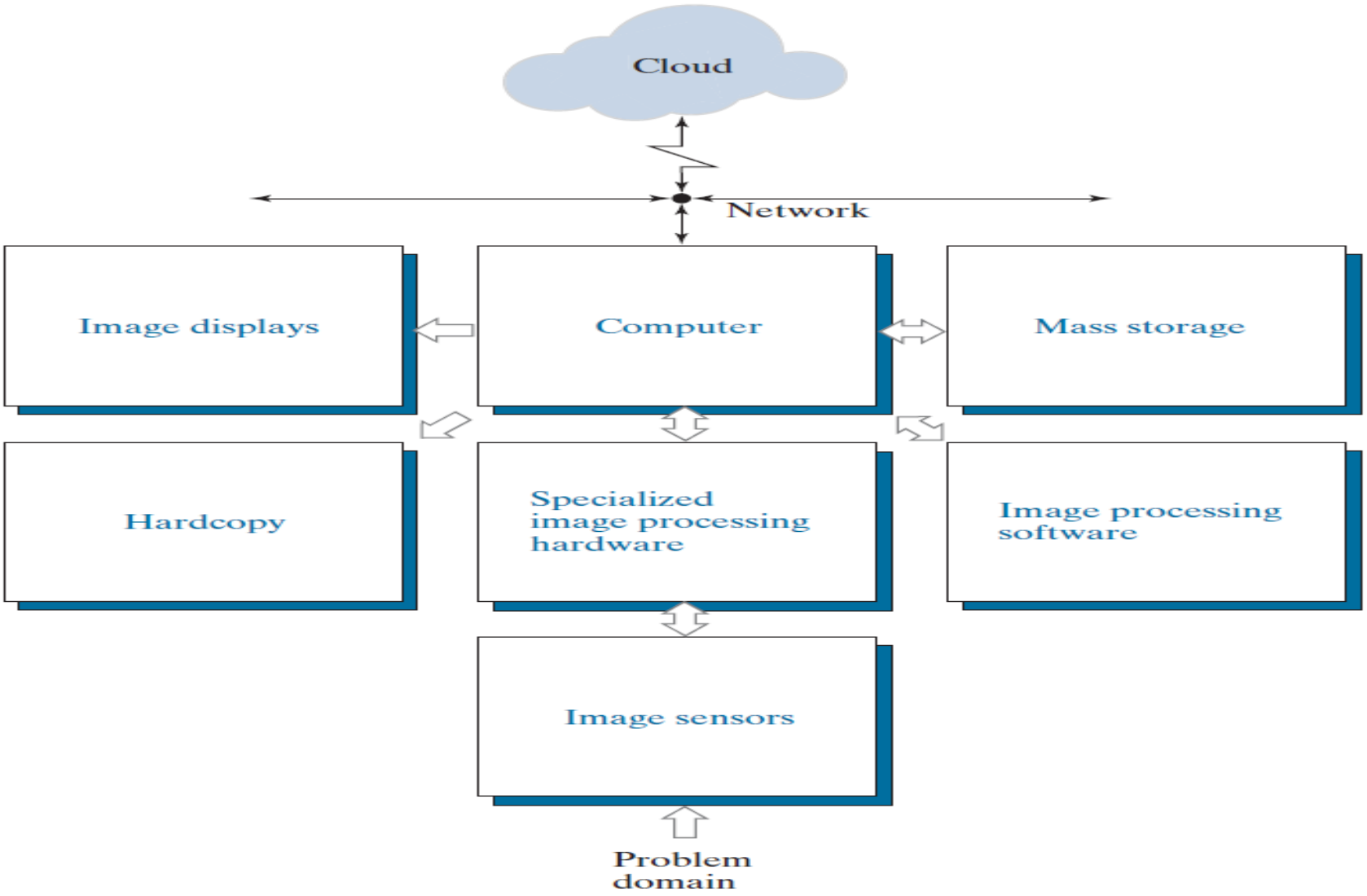


- Mid level processing
  - Input image
  - Output Attributes(edges, contours)
    - Example: object segmentation

- High level processing
  - Input attributes
  - Output understanding
    - Example: scene understanding



Basic components of any DIP system



OpenCv

- OpenCv stands for open source library for image processing and computer vision.
- OpenCv supports Windows, Linux, Android and Mac OS.
- OpenCv supports a wide variety of programming languages such as C++, Python, Java, etc..
- Primary interface of OpenCV is in C++
- **Opencv 4.3.0 Download link→**

https://sourceforge.net/projects/opencvlibrary/files/4.3.0/opencv-4.3.0-vc14_vc15.exe/download

```
#include<opencv2/opencv.hpp>
#include"iostream"
using namespace std;
using namespace cv;

int main()
{
	Mat image, gray_image;
	image = imread("C:\\Users\\Public\\Pictures\\Sample Pictures\\xx.jpg");
		//image = imread("xx.jpg");
	//Check for invalid input
	if (image.empty())
	{
		cout << "no image" << endl;
		system("pause");
		return -1;
	}
	namedWindow("RGB image", 0);
	imshow("RGB image", image);
	waitKey(0);

}
```




Thank you!